

Sheet 2 Lab Report

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1 Problem4

Q1 : Retrieve the name of students whose name start with 'S' and contains 'r' as the second last character

```
SELECT
    name
FROM
    students
WHERE
    TRIM(name) LIKE 'S%r_';
```

Output:

name
Swati Mitra

Q2 : Retrieve the name of the youngest student(s) from the ‘CST’ department along with the total marks obtained by him (them).

```
SELECT
    students.name,
    SUM(crs_regd.marks) as total_marks,
    AGE(CURRENT_DATE, students.bdate) as student_age
FROM
    students
JOIN crs_regd ON crs_regd.crs_rollno = students.rollno
WHERE
    students.deptcode = 'CST'
    AND AGE(CURRENT_DATE, students.bdate) = (
        SELECT
            MIN(AGE(CURRENT_DATE, bdate))
        FROM
            students
        WHERE
            deptcode = 'CST'
    )
GROUP BY
    students.rollno,
    students.name,
    students.bdate;
```

Output:

name	total_marks	student_age
Anjali Verma	176.00	30 years 4 mons 20 days

Q3: Find age of all students

```
SELECT
    name,
    rollno,
    AGE (CURRENT_DATE, bdate) as age
FROM
    students;
```

Output:

name	rollno	age
Riddhi Agarwal	92005001	31 years 5 mons 10 days
Priya Verma	92005002	30 years 1 mon 3 days
Rahul Das	92005003	30 years 9 mons 15 days
Sneha Gupta	92005004	30 years 7 mons 20 days
Ravi Kumar	92005010	31 years 2 mons 7 days
Neha Patil	92005011	29 years 10 mons 26 days
Vikram Singh	92005012	30 years 8 mons
Ananya Roy	92005013	30 years 4 mons 11 days
Suresh Nair	92005100	31 years 17 days
Kavita Jain	92005101	30 years 5 mons 28 days
Deepak Mehta	92005102	31 years 3 mons 5 days
Pooja Reddy	92005103	29 years 10 mons 13 days
Arun Bhat	92005104	31 years 6 mons 26 days
Meera Iyer	92005105	30 years 2 mons 20 days
Kiran Desai	92005106	30 years 11 mons 8 days
Aarav Sharma	92005201	30 years 3 mons 13 days
Diya Patel	92005202	31 years 5 days
Kabir Mehta	92005203	30 years 5 mons 14 days
Ishita Sen	92005204	30 years 8 mons 24 days
Rohan Bose	92005205	30 years 7 mons 3 days
Tanya Roy	92005206	30 years 10 mons 15 days
Arjun Singh	92005301	31 years 6 mons 11 days
Priya Das	92005302	30 years 1 mon 16 days
Karan Sharma	92005303	30 years 9 mons 5 days
Anjali Verma	92005304	30 years 4 mons 20 days
Rahul Gupta	92005305	30 years 11 mons 13 days
Sneha Patel	92005306	30 years 7 mons
Vikash Kumar	92005307	31 years 1 mon 26 days
Swati Mitra	92005405	30 years 11 days

2 Problem5

Q1 : Retrieve the name of the student(s) who obtained second highest marks in 'DBMS'.

```
SELECT
    s1.name
FROM
    crs_regd cr1
    JOIN students s1 ON s1.rollno = cr1.crs_rollno
    JOIN crs_offrd co ON co.crs_code = cr1.crs_cd
WHERE
    TRIM(co.crs_name) = 'DBMS'
    AND 1 = (
        SELECT
            COUNT(DISTINCT cr2.marks)
        FROM
            crs_regd cr2
            JOIN crs_offrd co2 ON co2.crs_code = cr2.crs_cd
        WHERE
            TRIM(co2.crs_name) = 'DBMS'
            AND cr2.marks > cr1.marks
    );
```

Output:

name
Riddhi Agarwal

Q2: Find out the differences between the highest and lowest marks obtained in each subject

```
SELECT
    co.crs_name ,
    MAX(marks) - MIN(marks) as marks_Difference
FROM
    crs_offrd co
    JOIN crs_regd cr ON cr.crs_cd = co.crs_code
GROUP BY
    crs_name;
```

Output:

crs_name	marks_difference
Thermodynamics	6.00
Metallurgy	0.00
DBMS	62.00
Chemistry	28.00
Software Engg.	0.00
Aerodynamics	0.00
Circuit Theory	12.00
Physics	35.00
Data Structures	6.00
Advanced Algorithms	0.00
OS	16.00
MIS	9.00
Basic Architecture	0.00
Signal Processing	25.00

Q3 : Assuming the existence of several interdepartmental courses, retrieve the name of the student(s) who is(are) studying under at least one faculty from each department.

```
SELECT
    name,
    rollno
FROM
    students
JOIN crs_regd ON crs_regd.crs_rollno = students.rollno
JOIN crs_offrd ON crs_regd.crs_cd = crs_offrd.crs_code
JOIN faculty ON faculty.fac_code = crs_offrd.crs_fac_cd
GROUP BY
    name,
    rollno
HAVING
    COUNT(DISTINCT TRIM(faculty.fac_dept)) = (
        SELECT
            COUNT(DISTINCT TRIM(deptcode))
        FROM
            depts
    );
```

Output:

name	rollno
Riddhi Agarwal	92005001

Q4: Assuming the existence of several interdepartmental courses, retrieve the name of the student(s) who is(are) studying under the faculties only from his(their) own department.

```
SELECT
    name,
    rollno
FROM
    students
JOIN crs_regd ON crs_regd.crs_rollno = students.rollno
JOIN crs_offrd ON crs_regd.crs_cd = crs_offrd.crs_code
JOIN faculty ON TRIM(faculty.fac_code) = TRIM(crs_offrd.crs_fac_cd)
GROUP BY
    name,
    rollno,
    deptcode
HAVING
    COUNT(DISTINCT TRIM(faculty.fac_dept)) = 1
    AND MAX(TRIM(faculty.fac_dept)) = TRIM(students.deptcode)
    AND COUNT(DISTINCT TRIM(faculty.fac_code)) = (
        SELECT
            COUNT(DISTINCT TRIM(faculty.fac_code))
        FROM
            faculty
        WHERE
            TRIM(fac_dept) = TRIM(deptcode)
    );
```

Output:

name	rollno
Priya Verma	92005002

3 Problem6

Q1: Display the highest parent incomes, in descending order, for each department excluding 'ARCH' such that only those highest parent incomes will appear that are below 12,000.

```
SELECT
    s1.rollno ,
    s1.name ,
    s1.deptcode ,
    s1.parent_inc
FROM
    students s1
WHERE
    deptcode != 'ARCH'
    AND parent_inc < 12000
    AND s1.parent_inc = (
        SELECT
            MAX(s2.parent_inc)
        FROM
            students s2
        WHERE
            s1.deptcode = s2.deptcode
            AND s2.deptcode != 'ARCH'
            AND s2.parent_inc < 12000
    )
ORDER BY
    s1.parent_inc DESC;
```

Output:

rollno	name	deptcode	parent_inc
92005205	Rohan Bose	IT	11800.0
92005202	Diya Patel	CSE	11500.0
92005204	Ishita Sen	ELE	9500.0

Q2: Retrieve the 5th largest parental income for hostel no. 5

```
SELECT
    s1.rollno,
    s1.name,
    s1.parent_inc
FROM
    students s1
WHERE
    4 = (
        SELECT
            COUNT(DISTINCT s2.parent_inc)
        FROM
            students s2
        WHERE
            s2.parent_inc > s1.parent_inc
    );
```

Output:

rollno	name	parent_inc
92005101	Kavita Jain	650000.0

Q3: Find the roll number of the students from each department who obtained highest total marks in their own department.

```
SELECT
    s1.name ,
    s1.rollno ,
    s1.deptcode ,
    SUM(cr1.marks) as total_marks
FROM
    students s1
    JOIN crs_regd cr1 ON cr1.crs_rollno = s1.rollno
GROUP BY
    s1.name ,
    s1.rollno ,
    s1.deptcode
HAVING
    SUM(cr1.marks) = (
        SELECT
            (MAX(dept_totals.total))
        FROM
            (
                SELECT
                    s2.deptcode ,
                    SUM(cr2.marks) as total
                FROM
                    students s2
                    JOIN crs_regd cr2 ON cr2.crs_rollno = s2.rollno
                GROUP BY
                    s2.rollno ,
                    s2.deptcode
            ) as dept_totals
        WHERE
            dept_totals.deptcode = s1.deptcode
    )
ORDER BY
    total_marks;
```

Output:

name	rollno	deptcode	total_marks
Kiran Desai	92005106	MME	71.00
Ravi Kumar	92005010	ELE	125.00
Arun Bhat	92005104	IT	134.00
Pooja Reddy	92005103	EE	175.00
Deepak Mehta	92005102	ETC	187.00
Suresh Nair	92005100	CST	295.00
Riddhi Agarwal	92005001	CSE	1101.00